Abstract

Objectives: Quantitative spinal cord MRI biomarkers vary systematically with scanner vendor, which limits multi-centre pooling, yet no harmonisation benchmark exists for spinal cord data.

Methods: Eight biomarkers (T2w-CSA, GM-CSA, MTR, MTsat, FA, MD, AD, RD) were extracted at C2–C3 from the open spine-generic dataset (267 participants, 43 sites, three vendors; site and vendor perfectly nested). Five post-hoc methods—ComBat, ComBat-joint, CovBat, RELIEF and a mixed-effects/fixed-effects hybrid (LME/FE)—were benchmarked on four endpoints in healthy-control ($N = 188$) and pathology-inclusive ($N = 246$) cohorts: vendor-effect removal, biological-signal recovery, subject-level preservation, and machine-learnability of the vendor signature. Leave-one-vendor-out validation tested generalisation to unseen manufacturers.

Results: Baseline vendor effects were large ($\eta^2 = 0.31$; PERMANOVA $R^2 = 0.315$, $p = 1 \times 10^{-4}$). All methods reduced the vendor variance fraction to 2.5–7.9 $\times 10^{-4}$, but this criterion was nearly degenerate because vendor is by construction the harmonisation variable. In leakage-free evaluation, machine detection of vendor identity fell from an AUC of about 0.99 to 0.63–0.81, while pathology classification improved from 0.52 to 0.62. Biological-signal recovery was selective: age associations absent at baseline became significant in diffusion metrics after harmonisation ($p \leq 0.0019$). Methods diverged on subject-level preservation ($r = 0.78$–0.82), forming a Pareto frontier. Only ComBat and ComBat-joint generalised to an unseen vendor without significant degradation (PERMANOVA $p \geq 0.97$); RELIEF and LME/FE did not ($\eta^2 \approx 0.2$, $p < 0.01$). Three structural constraints—vendor-site nesting, mixed-effects fixed-effects fallback, and latent-factor inactivation—converge on one limitation: only three vendors in a perfectly nested design.

Conclusion: Harmonisation may be appropriate for pooled multi-vendor spinal cord quantitative MRI datasets of comparable structure. ComBat-joint, which harmonises all eight biomarkers jointly, provided the most stable generalisation to unseen vendors and the best subject-level preservation among ComBat-family methods, and may be appropriate for broad deployment.

Key Points:

1. Scanner vendor effects in spinal cord qMRI are quantitatively large in our sample ($\eta^2 = 0.31$, approximately 155× the variance from age and 16× that from sex), motivating harmonisation before multi-centre pooling.
2. ComBat-family harmonisation removes most machine-detectable vendor information (leakage-free AUC 0.99 to ≈0.70 for ComBat and ComBat-joint; chance 0.50) while unmasking age- and pathology-related biological signal previously obscured by vendor variance.

3. Leave-one-vendor-out validation reveals that latent-structure (RELIEF) and random-effects (mixed-model) approaches fail to generalise to new scanner vendors; among the methods that did transfer, joint ComBat estimation was the most stable (LOVO/internal η^2 ratio 0.9 vs. 6.0 for per-metric ComBat), supporting ComBat-joint for multi-centre consortia.

Keywords: Spinal cord; Quantitative MRI; Harmonisation; ComBat; Multicentre study; Reproducibility; Scanner vendor effects

Abbreviations: AD = axial diffusivity; ComBat = Combatting batch effects; CovBat = Covariance Batch harmonisation; CSA = cross-sectional area; DCM = degenerative cervical myelopathy; FA = fractional anisotropy; HC = healthy control; ICA = independent component analysis; LME/FE = linear mixed-effects / fixed-effects hybrid; LOVO = leave-one-vendor-out; MD = mean diffusivity; MT = magnetisation transfer; MTR = magnetisation transfer ratio; MTsat = magnetisation transfer saturation; PERMANOVA = permutational multivariate analysis of variance; qMRI = quantitative magnetic resonance imaging; RD = radial diffusivity; RELIEF = REmoval of Latent Inter-scanner Effects through Factorization

1 Introduction

Quantitative MRI (qMRI) biomarkers of the spinal cord—cross-sectional area (CSA), magnetisation transfer ratio (MTR), magnetisation transfer saturation (MTsat), and diffusion tensor metrics (FA, MD, AD, RD)—provide non-invasive windows into atrophy, demyelination, and axonal integrity in multiple sclerosis, degenerative cervical myelopathy, and spinal cord injury [1–3]. Multi-centre consortia are increasingly adopting these biomarkers, but systematic vendor-dependent variability limits cross-site pooling [4,5]. A recent pan-European survey by the European Society of Radiology documented marked heterogeneity in MRI quality assurance and control practices across 37 countries [6], underscoring the urgency of robust harmonisation.

Even under standardised acquisition protocols, differences in gradient hardware, RF coil geometry, and vendor-specific reconstruction pipelines introduce offsets that often exceed the biological effect sizes of interest [4,7]. Statistical post-hoc harmonisation methods—ComBat [8–10], CovBat [11], RELIEF [12], and linear mixed-effects models (LME) [13]—have been extensively benchmarked in brain imaging and reviewed in the broader context of multicentre radiomics [14] but remain uncharacterised for spinal cord qMRI. Fortin et al. first applied ComBat to multi-site DTI [10], establishing a framework later extended to cortical thickness [9], longitudinal harmonisation [15], covariance harmonisation [11], latent-factor removal [12], and out-of-sample harmonisation for new

sites [16]. Middleton et al. demonstrated ComBat's effectiveness for spinal cord DTI [17], and Jodoin et al. provided deployment guidance for diffusion MRI [18], but no study has systematically compared the full ComBat family across a comprehensive spinal cord biomarker panel using unified multi-endpoint evaluation.

The spinal cord presents distinct harmonisation challenges: small per-site sample sizes in the complete-case sample (HC: median 5 subjects per site, range 2–8; full dataset: median 5, range 1–8) challenge empirical Bayes shrinkage; strong inter-metric correlations (diffusion metrics are intrinsically coupled; MTR and MTsat share acquisition-dependent variance) motivate multivariate evaluation; and the clinical need to preserve disease-related variance creates a tension not previously quantified.

Here we benchmark five methods on eight spinal cord qMRI biomarkers using four orthogonal endpoints: vendor-effect removal, biological-signal recovery, subject-level preservation, and machine-learnability of the vendor signature. We extend prior work through: (i) formal comparison of the full ComBat family (including covariance and latent-factor extensions); (ii) multivariate evaluation via PERMANOVA; (iii) a Pareto-frontier analysis of the decontamination–preservation trade-off; and (iv) leave-one-vendor-out external validation to test generalisability.

2 Materials and Methods

2.1 Dataset and Cohorts

The spine-generic multi-subject dataset (release r20231212) [1] comprises 267 subjects scanned at 43 sites across three vendors (GE Healthcare: 37 subjects at 6 sites; Philips: 50 subjects at 8 sites; Siemens Healthineers: 180 subjects at 29 sites) using a standardised consensus protocol [4]; the dataset composition and the eight-biomarker panel are illustrated in Figure 1a,b. Site and vendor are perfectly nested (each site operates a single scanner from one vendor); reported vendor effects thus represent the composite "vendor-in-site" effect. The complete-case analysis sample comprised 40 sites (GE 5, Philips 8, Siemens 27). Demographics were balanced across vendors in the complete-case analysis sample (Table 1; all $p \geq 0.203$).

Table 1: Demographic and acquisition characteristics by vendor (HC complete-case analysis sample, $N = 188$). Continuous variables: mean ± SD, one-way ANOVA. Categorical variables: count (%), χ^2 test. Site and vendor are perfectly nested (40 sites across 3 vendors in the analysis sample; 43 sites in the full dataset).

Characteristic	GE	Philips	Siemens	Total	p-value
N	25	33	130	188	—
Sites (n)	5	8	27	40	—
Subjects per site, median [range]	5 [2–8]	4 [2–6]	5 [2–8]	5 [2–8]	—
Age, years (mean ± SD)	27.0 ± 4.9	27.9 ± 4.1	29.0 ± 6.0	28.5 ± 5.6	0.203
Age range	19–39	23–38	20–52	19–52	—
Sex = F	13 (52.0%)	18 (54.5%)	60 (46.2%)	91 (48.4%)	0.640
Sex = M	12 (48.0%)	15 (45.5%)	70 (53.8%)	97 (51.6%)	—
Height, cm	170.1 ± 8.7	171.3 ± 9.9	173.2 ± 10.1	172.5 ± 9.9	0.277
Weight, kg	66.4 ± 13.6	66.5 ± 13.6	67.9 ± 13.1	67.5 ± 13.2	0.781
Scanner models (n)	3	4	5	12	—

Two pre-specified cohorts were analysed: HC ($N = 188$ complete-case healthy controls) isolates pure vendor-driven variance; ALL ($N = 246$ complete-case, including 56 mild cervical compression and 2 DCM) approximates real-world deployment. Pathology distribution was balanced across vendors (χ^2 $p = 0.661$). Detailed cohort flow and missingness patterns are provided in Supplementary Material S1.

2.2 Biomarker Extraction and Processing

Eight biomarkers spanning four acquisition families were extracted at the C2–C3 level (Figure 1b) using the Spinal Cord Toolbox (SCT v7.4) [19]: T2w-CSA (T2-weighted structural), GM-CSA ($T2^*$-weighted), MTR and MTsat (magnetisation transfer), and FA, MD, AD, RD (diffusion tensor imaging). Processing followed the spine-generic pipeline documentation. Automated quality-control metrics excluded acquisitions with segmentation or model-fit failures (see Supplementary Material S2).

2.3 Harmonisation Methods

Five methods were evaluated (full mathematical details in Supplementary Material S3). In all methods, scanner vendor (GE/Philips/Siemens, three levels) served as the batch variable; because sites are perfectly nested within vendors, vendor-level and site-level offsets are statistically confounded, and the composite "vendor-in-site" effect is the harmonisation target:

ComBat [8–10]: Per-metric location-and-scale adjustment with empirical Bayes shrinkage on vendor-specific parameters. Each biomarker harmonised independently.

ComBat-joint [9]: All eight biomarkers harmonised jointly (joint empirical Bayes), preserving cross-metric covariance structure.

CovBat [11]: ComBat-joint followed by principal component (PC) correction—residuals projected onto PCs, each PC harmonised independently, then projected back. PCs explaining at least 95% of the residual variance were retained: $K = 5$ of 8 (cumulative variance 95.9% in the HC cohort, 95.6% in the ALL cohort; the first three PCs accounted for 85.6% and 84.9%, respectively).

RELIEF [12]: ComBat-joint followed by independent component analysis (ICA) of residuals. Our implementation is a residual-ICA variant of RELIEF (REmoval of Latent Inter-scanner Effects through Factorization), rather than the linked matrix factorisation of the original report; this distinction is revisited in the Limitations. Components with significant vendor association (Benjamini-Hochberg FDR-controlled ANOVA, $q < 0.05$) are zeroed. In the present dataset, no component reached significance (all $q \geq 0.99$), rendering the pipeline functionally equivalent to ComBat-joint with an additional PCA/ICA round-trip.

LME/FE: Vendor as a random intercept with age and sex fixed effects. When the random-effects covariance matrix was singular (expected with only 3 vendor levels—well below the recommended $k \geq 5$ for stable REML estimation), the model fell back to ordinary least squares centring (9 of 16 biomarker×cohort combinations, 56%; see Results). Only MD and AD achieved stable LME fits in both cohorts. We label this hybrid approach "LME/FE" to acknowledge the dual mechanism.

2.4 Evaluation Endpoints

Four orthogonal endpoints were evaluated per method per cohort (detailed in Supplementary Material S4):

Vendor-effect removal: Univariate η^2 (ANOVA), intraclass correlation coefficient (ICC), and multivariate PERMANOVA R^2 (Euclidean distance, 9,999 permutations) [20]. All five methods and the unharmonised baseline—including ComBat and LME/FE, which were refitted on the unified complete-case sample—were evaluated on the same subjects, with per-dimension z-scoring of biomarkers before PERMANOVA. Consequently, the multivariate R^2 equals the mean of the per-metric η^2 for every method (a mathematical identity of z-scored Euclidean PERMANOVA), so univariate and multivariate summaries are directly comparable throughout.

Biological-signal recovery: Type-II semi-partial R^2 for age and sex from linear models.

Subject-level preservation: Pearson r between pre- and post-harmonisation per-subject biomarker vectors.

Machine-learnability of the vendor signature: Following the site-detection

paradigm of Chen et al. [11], a random forest (500 trees, minimum leaf size 5, balanced class weights) was trained to classify scanner manufacturer (Siemens/GE/Philips) from the 8-biomarker profile of each subject, using repeated stratified 5-fold cross-validation (20 repeats). To prevent information leakage, each harmonisation model was refitted on the training folds only and applied to the held-out fold within every cross-validation split; because scanner vendor is observable at deployment, vendor labels were available to the harmonisation step on training and test folds alike, while the random forest itself remained blind to vendor labels. The macro one-vs-rest AUC quantifies residual vendor information detectable by machine learning, a stricter criterion than univariate η^2 because nonlinear combinations of features can encode vendor structure invisible to marginal statistics. Biological-signal preservation was indexed by the same classifier applied to pathology (healthy control vs disease, ALL cohort, $N = 246$ complete cases) and sex labels. A shuffled-label permutation reference calibrated the chance level.

2.5 Leave-One-Vendor-Out External Validation

For each method, harmonisation models were trained on subjects from two vendors and applied to the held-out third vendor in three rotation folds. Test-vendor transformation used z-score distribution matching (ComBat, ComBat-joint) or projection onto training PCA/ICA space (CovBat, RELIEF). LME/FE assigned offset=0 to unseen vendors. Three of the four endpoints (vendor-effect removal, biological-signal recovery, and subject-level preservation; machine-learnability and the ALL-cohort pathology/sex preservation indices were not re-evaluated) were evaluated after pooling the harmonised training-vendor and held-out test-vendor data within each fold (Supplementary Material S5). For ComBat, ComBat-joint, and LME/FE, the test-vendor transformation applies a per-biomarker affine (location–scale) map, so the test-vendor pre/post correlation equals 1.0 by construction. For CovBat and RELIEF, in contrast, projection onto the training PCA/ICA basis mixes biomarker dimensions, and the test-vendor pre/post correlation fell below 1 (fold-wise range $r = 0.85$–0.97 across the three held-out vendors; fold means 0.90 for CovBat and 0.95 for RELIEF). This indicates genuine distortion of held-out-vendor subject rankings. LOVO preservation statistics for the first group therefore reflect the pooled sample rather than the unseen vendor in isolation. Similarly, the LOVO/internal η^2 ratio below 1 for ComBat-joint reflects the pooled evaluation sample rather than genuine improvement on unseen vendors.

2.6 Supplementary Analyses

Design C reduced-imbalance subsample analysis restricted Siemens to 40 randomly drawn subjects per seed (HC cohort, retaining all GE and Philips subjects; $n \approx 98$ per subset; residual vendor imbalance max / min = 1.60 vs. 5.2 in the full analysis; 20 random seeds),

with all methods refitted per subset using the same canonical implementations as the main analysis, to isolate the effect of vendor imbalance from method performance. A true-equal-n sensitivity analysis (25 subjects per vendor, $n = 75$, imbalance ratio = 1.00) is reported in Supplementary Material S6.

2.7 Uncertainty Quantification

Ninety-five per cent confidence intervals for key point estimates were obtained using a stratified nonparametric bootstrap (B = 2,000), resampling subjects with replacement within each scanner-vendor stratum to preserve group proportions. Percentile intervals are reported for $r_{\text{pre,post}}$ and for the baseline vendor effect size. Because marginal CIs for $r_{\text{pre,post}}$ overlap across methods, method pairs were additionally compared with a within-resample paired bootstrap: for each bootstrap sample, all methods' r values were recomputed and their differences tested against zero, which removes between-resample variance and directly targets the pairwise contrast. Bootstrap intervals are not reported for post-harmonisation η^2 or PERMANOVA R^2 because harmonisation methods optimise on the same data, rendering subject-level resampling intervals unreliable for these near-degenerate statistics.

2.8 Reporting Guidelines and Missing Data

This study follows STROBE reporting guidance for cross-sectional studies where applicable. The spine-generic dataset is a curated benchmark with pre-processed, quality-controlled biomarker measurements. Subjects with missing biomarker data were excluded via complete-case analysis (HC: N = 188/203, 7.4%; ALL: N = 246/267, 7.9%); missingness was attributable to acquisition-level exclusions (motion artefacts, segmentation failures, and incomplete sequences) as detailed in Supplementary Material S1. Sensitivity to missing-data assumptions was assessed through the ALL-cohort analysis, which preserves a broader subject spectrum.

2.9 Use of AI-Assisted Tools

An AI-based writing assistant (WorkBuddy, based on a large language model) was used to improve English language clarity, grammar, and conciseness of author-written text, and to assist with the structural organisation of the manuscript and the author response documents. No AI tool was used for study conception or design, data acquisition, data analysis, statistical computation, generation of figures, or interpretation of results; all numerical results, tables, and figures were produced by the authors using standard scientific software and independently verified against the underlying data. All AI-assisted text was

reviewed, fact-checked, and edited by the authors, who take full responsibility for the content of the manuscript.

3 Results

3.1 Baseline Vendor Effects

At baseline, vendor explained 31% of univariate biomarker variance (mean $\eta^2 = 0.315$, 95% CI [0.273, 0.381], stratified bootstrap; range 0.05–0.70 across metrics) and 31% of multivariate structure (PERMANOVA $R^2 = 0.315$, $p = 1 \times 10^{-4}$, the permutation lower bound with 9,999 permutations; HC cohort). These effects exceeded age-related variance by approximately two orders of magnitude and sex-related variance by one order. AD showed the largest vendor effect ($\eta^2_{\mathrm{HC}} = 0.70$), followed by MD (0.46) and MTsat (0.34); among structural metrics, GM-CSA showed the strongest vendor discrepancy ($\eta^2 = 0.18$, Philips scanners systematically lower; Supplementary Fig. S3). In the ALL cohort, baseline vendor effects were comparable or larger ($\eta^2 = 0.336$; PERMANOVA $R^2 = 0.336$).

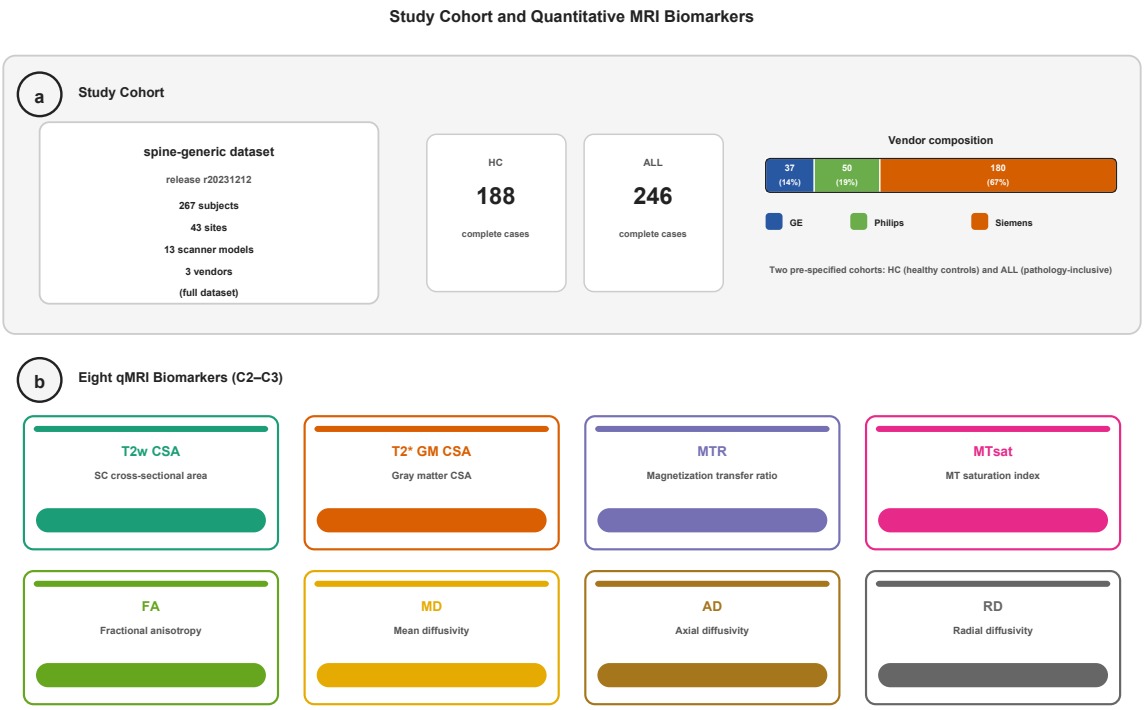


Figure 1: Overview of the study dataset and biomarker panel. **(a)** Study cohort: the spine-generic multi-subject dataset (release r20231212) [1] comprising 267 subjects scanned at 43 sites on 13 scanner models from three vendors (GE: 37 subjects at 6 sites; Philips: 50 at 8; Siemens: 180 at 29), with the two pre-specified analysis cohorts (HC, $N = 188$ complete cases; ALL, $N = 246$ complete cases). **(b)** The eight quantitative MRI biomarkers extracted at the C2–C3 level, spanning four acquisition families: T2w cross-sectional area (T2w-CSA) from T2-weighted structural imaging; grey matter CSA (GM-CSA) from $T2^*$-weighted imaging; magnetisation transfer ratio (MTR) and saturation (MTsat); and fractional anisotropy (FA), mean (MD), axial (AD), and radial (RD) diffusivity from diffusion tensor imaging.

3.2 Vendor-Effect Removal

All five methods reduced vendor η^2 to between 2.5×10^{-4} (CovBat, HC) and 7.9×10^{-4} (LME/FE, HC; mean 4.9×10^{-4} across methods; Figure 2a,b, Table 2), corresponding to a 400- to 1,260-fold reduction from the baseline mean of 0.315 (mean ≈ 2.7 orders of magnitude across methods). For reference, the η^2 expected under the null hypothesis of no vendor effect is $(k-1)/(N-1) = 2/187 \approx 0.011$; post-harmonisation values lie far below this chance level, so the effective reduction relative to chance expectation is 92.6–97.6% across methods (LME/FE lowest at 92.6%, CovBat highest at 97.6%). Multivariate vendor structure became indistinguishable from chance (all PERMANOVA $p = 1.0$ in both cohorts). These near-zero values should, however, be read as a process check rather than an independent validation: vendor is the harmonisation batch variable, and each

method explicitly removes vendor location and scale, so a correctly implemented pipeline is expected to drive η^2 toward zero by construction. Machine-learnability (below) provides the stricter, non-degenerate criterion.

Covariance structure was displaced from the original to a comparable degree by all methods (Supplementary Fig. S4). Expressed relative to the total Frobenius norm of the original cross-metric correlation matrix ($\|R_{\text{orig}}\|_F = 4.08$ in HC, 4.07 in ALL), post-harmonisation correlation matrices lay 37.4–41.3% away from the original in the HC cohort (ComBat 37.4%, LME/FE 38.3%, ComBat-joint 38.5%, RELIEF 40.3%, CovBat 41.3%; corresponding range 39.8–43.0% in ALL: ComBat 39.8%, LME/FE 40.5%, ComBat-joint 40.5%, RELIEF 42.3%, CovBat 43.0%). No method therefore preserved the covariance structure in absolute terms. Between-method Frobenius distances ranged from 0.041 (ComBat vs. ComBat-joint) to 0.094 (CovBat vs. RELIEF), 16–42× smaller than the typical method-vs-original displacement (1.5–1.7). CovBat and RELIEF yielded near-identical correlation matrices to each other (Frobenius distance 0.09 in HC, 0.08 in ALL), as did ComBat and ComBat-joint (0.11 in HC, 0.10 in ALL); the location–scale methods remained marginally closer to the original than the covariance-projection methods.

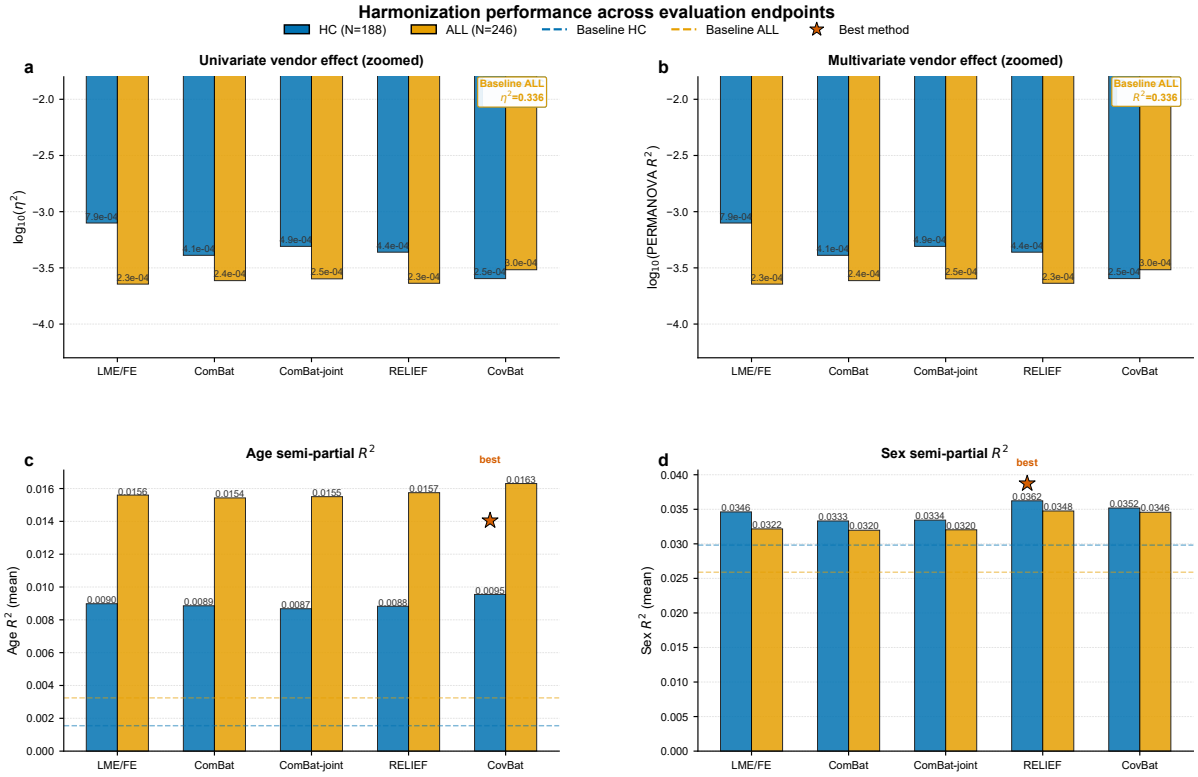


Figure 2: Harmonisation performance by method and endpoint (healthy control cohort; the baseline row shows the Original, unharmonised data). **(a)** Vendor effect size (η^2, eta-squared from one-way ANOVA) before and after harmonisation, shown on logarithmic scale. **(b)** Multivariate vendor structure (PERMANOVA R^2, permutational multivariate analysis of variance) on logarithmic scale. **(c)** Age-related biological signal (semi-partial R^2) before and after harmonisation. **(d)** Sex-related biological signal (semi-partial R^2) before and after harmonisation.

Table 2: Harmonisation performance summary (HC cohort, complete-case $N = 188$). All methods were fitted and evaluated on the same unified complete-case sample. Values are means across 8 biomarkers. η^2_{vendor}: post-harmonisation vendor ANOVA effect size; $R^2_{\text{PERMANOVA}}$: multivariate vendor structure (equals the mean per-metric η^2 by the z-score Euclidean identity); $r_{\text{pre,post}}$: subject-ranking preservation; Age R^2: semi-partial age variance explained. LOVO columns show leave-one-vendor-out external validation (PERMANOVA R^2 with z-scored Euclidean distance on the pooled within-fold sample, mean across 3 held-out-vendor folds); LOVO ratio: LOVO η^2 relative to internal η^2. Values in parentheses are 95% percentile bootstrap CIs ($B = 2{,}000$, stratified by vendor).

Method	η^2_{vendor}	$R^2_{\text{PERMANOVA}}$	$r_{\text{pre,post}}$ (95% CI)	Age R^2	LOVO η^2	LOVO ratio
Original (baseline)	3.15×10^{-1}	0.315	1.000	0.002	—	—
ComBat	4.09×10^{-4}	0.0004	0.789 (0.741–0.825)	0.009	2.46×10^{-3}	6.0×
ComBat-joint	4.91×10^{-4}	0.0005	0.806 (0.760–0.839)	0.009	4.42×10^{-4}	0.9×
CovBat	2.54×10^{-4}	0.0003	0.782 (0.735–0.816)	0.010	1.01×10^{-3}	4.0×
RELIEF	4.36×10^{-4}	0.0004	0.791 (0.743–0.824)	0.009	1.87×10^{-1}	428 ×
LME/FE	7.93×10^{-4}	0.0008	0.816 (0.771–0.848)	0.009	2.01×10^{-1}	253 ×

3.3 Machine-Learnability of the Vendor Signature

Unharmonised biomarker profiles allowed near-perfect machine detection of scanner manufacturer (macro one-vs-rest AUC $= 0.986 \pm 0.002$ in the ALL cohort; 0.983 ± 0.002 in HC), confirming that the vendor signature dominates the multivariate feature space. After harmonisation—refitted within each cross-validation fold to prevent leakage of harmonisation parameters (Methods)—detectability decreased substantially but remained above chance for every method (permutation reference 0.495 ± 0.038): CovBat 0.629 ± 0.020, RELIEF 0.690 ± 0.017, ComBat-joint 0.704 ± 0.018, ComBat 0.706 ± 0.016, and LME/FE 0.811 ± 0.015 (ALL cohort, $N = 246$; HC values were concordant: 0.670, 0.733, 0.756, 0.751, and 0.851, respectively). Expressed against the chance-adjusted detectability margin, location–scale harmonisation removed 36% (LME/FE) to 73% (CovBat) of the machine-detectable vendor information; the method ordering parallels correction depth, the FE fallback being a pure location shift without scale adjustment. Per-vendor detectability after ComBat ranged from 0.67 (Philips) to 0.80 (GE) (Supplementary Table S8). Complementarily, pathology classification improved from AUC 0.520 ± 0.024 (unharmonised) to 0.582–0.625 across methods (ComBat highest, 0.625 ± 0.018; point AUC 0.64, 95% CI 0.55–0.72; within-resample paired comparison vs. Original: mean ΔAUC +0.105, $p < 0.001$ for all five methods by paired t and Wilcoxon tests; the unpaired Hanley–McNeil test on point AUCs did not reach significance, $p = 0.07$, reflecting the power gained by fold-level pairing), and sex classification remained stable (0.646–0.670), indicating that harmonisation removed vendor confounding while unmasking—rather than distorting—biological signal.

3.4 Biological-Signal Recovery

Before harmonisation, no biomarker showed a significant age association in either cohort (HC: all $p \geq 0.34$; ALL: all $p \geq 0.19$). After harmonisation, significant age associations emerged selectively in the diffusion metrics: MD and AD in both cohorts (all methods; HC $p = 0.006$–0.049, ALL $p \leq 0.0019$) and additionally RD in ALL ($p \leq 0.007$); structural and magnetisation-transfer metrics remained non-significant (all $p \geq 0.18$). In mean terms, age R^2 increased 5.6–6.2-fold in HC (from 1.5×10^{-3} to 8.7×10^{-3}–9.5×10^{-3}) and 4.8–5.0-fold in ALL (from 3.2×10^{-3} to 1.5×10^{-2}–1.6×10^{-2}); these fold changes should be read alongside the absolute values and the per-metric significance above, since a ratio of two near-zero variances is itself unstable. Sex R^2 showed a similar pattern (1.1–1.3-fold increase). Absolute age effects were larger in ALL, consistent with greater biological heterogeneity, but remained below 2% of variance.

3.5 Subject-Level Preservation and the Pareto Frontier

Methods diverged on subject-ranking preservation (HC cohort, mean r across methods: LME/FE = 0.816, ComBat-joint = 0.806, RELIEF = 0.791, ComBat = 0.789, CovBat = 0.782). This ranking was highly concordant between cohorts (Spearman $\rho = 0.90$), with a single adjacent exchange of ranks (ComBat and RELIEF swapped ranks 3–4; ALL values: LME/FE 0.801, ComBat-joint 0.787, ComBat 0.774, RELIEF 0.772, CovBat 0.763). Although the marginal bootstrap 95% CIs for $r_{\text{pre,post}}$ overlap across methods (Table 2), a within-resample paired bootstrap ($B = 2{,}000$) confirmed that the preservation hierarchy is statistically reliable: 8 of 10 method pairs were significantly separated in HC (all $p \leq 0.003$) and 9 of 10 in ALL, with LME/FE significantly highest and CovBat significantly lowest (the only non-separated pairs were ComBat vs. CovBat and ComBat vs. RELIEF in HC, and ComBat vs. RELIEF in ALL; Supplementary Table S7).

In the HC cohort, a Pareto frontier emerged, with methods trading greater multivariate decontamination for lesser subject-level preservation (Figure 3). Because post-harmonisation PERMANOVA R^2 values are near-degenerate (Section 3.2), we repeated the frontier ordering using leakage-free vendor detectability (Section 3.3) as the decontamination axis: the corner solutions were preserved (CovBat: maximal removal; LME/FE: maximal preservation), and ComBat and ComBat-joint were indistinguishable on removal in HC (0.751 vs. 0.756, within cross-validation noise) with ComBat-joint showing higher preservation (0.806 vs. 0.789); in the pathology-inclusive cohort, ComBat-joint weakly dominated ComBat on both axes (lower detectability, 0.704 vs. 0.706, and higher preservation, 0.787 vs. 0.774), although the removal difference again lies within cross-validation noise. LME/FE's top rank partly reflects its minimal correction: the FE fallback applies a pure location shift without scale adjustment, leaving subject-level variance essentially untouched.

Under the reduced-imbalance subsample (Design C, $n \approx 98$ per seed, imbalance ratio 1.60 vs. 5.2 in the full analysis), the pooled-r preservation hierarchy was retained (LME/FE $r = 0.779 \pm 0.011$, ComBat-joint 0.773 ± 0.014, RELIEF 0.760 ± 0.015, CovBat 0.749 ± 0.016, ComBat 0.745 ± 0.015). The only rank change was the adjacent exchange of ComBat and CovBat, a pair not significantly separated in the main analysis. A true-equal-n sensitivity analysis (25 subjects per vendor, $n = 75$, ratio 1.00) gave concordant results (LME/FE 0.772 ± 0.015; ComBat-family 0.732–0.768).

Absolute values were lower than in the full cohort (0.782–0.816) because both subsample designs carried a higher baseline vendor-variance fraction: the measured baseline η^2 was 0.374 (range 0.343–0.401 across seeds) under reduced imbalance and 0.389 (range 0.337–0.431) under exact balance, versus 0.315 in the full analysis. Because pooled r after a location–scale correction cannot exceed $\sqrt{1-\eta^2}$, these baselines imply ceilings of approximately 0.79 and 0.78, respectively; every per-method mean fell below its design's ceiling, and the per-seed LME/FE estimate never exceeded the per-seed ceiling in any of the 20 seeds of either design (Supplementary Material S6). Within-vendor subject rankings were nonetheless preserved near-perfectly by every method in both subsample designs (mean within-vendor $r \geq 0.97$), indicating that the pooled-r gradient reflects differential removal of between-vendor variance rather than degradation of within-vendor subject rankings (Supplementary Material S6). The preservation hierarchy therefore persists under reduced imbalance and under exact balance, indicating that it tracks correction depth rather than the dataset's vendor imbalance; the Pareto frontier nonetheless remains a descriptive characterisation of this dataset's structure.

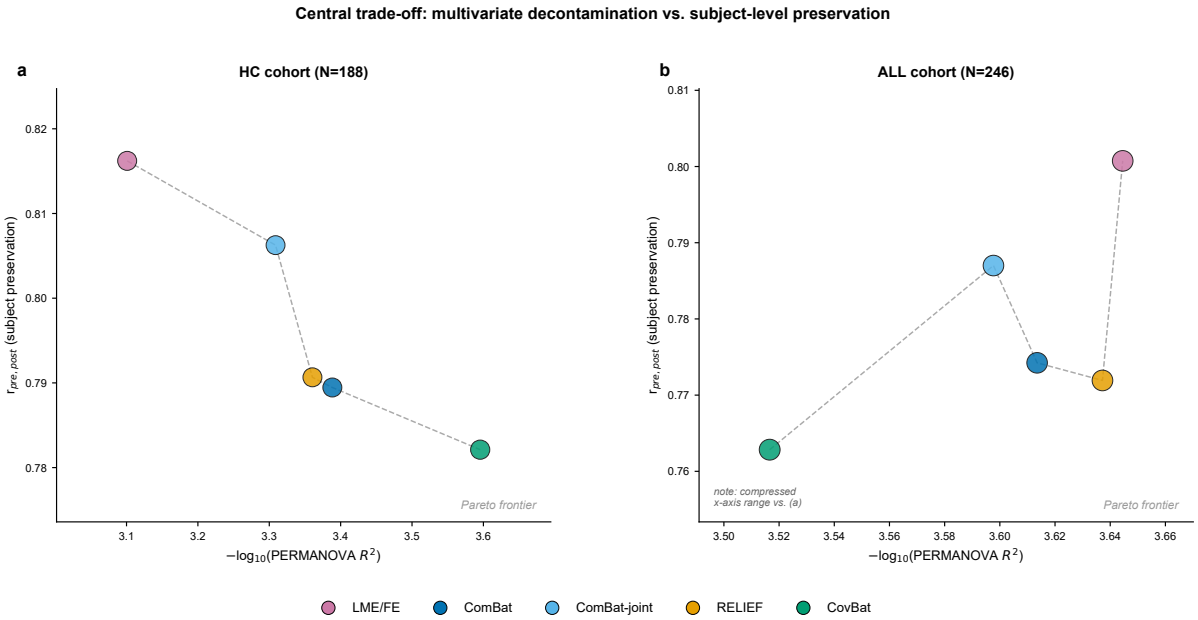


Figure 3: Pareto frontier showing the trade-off between multivariate vendor-effect removal (PERMANOVA R^2, permutational multivariate analysis of variance) and subject-level preservation (Pearson correlation $r_{\text{pre,post}}$ between pre- and post-harmonisation biomarker vectors). **(a)** Healthy control cohort ($N = 188$); **(b)** pathology-inclusive cohort ($N = 246$). A monotone frontier is present in the HC cohort; in the pathology-inclusive cohort the ordering is non-monotone (Section 3.5). Each point represents one harmonisation method. Lower right indicates maximal vendor-effect removal; upper left indicates maximal subject-level preservation. The x-axis range in **(b)** is compressed relative to **(a)** (flagged in the panel), so the apparent spread of methods on this axis should not be compared directly between panels.

3.6 LOVO External Validation

ComBat and ComBat-joint generalised to unseen manufacturers without significant degradation (PERMANOVA $p \geq 0.97$ and $p = 1.0$ across the three folds, respectively), although ComBat's point estimate rose to LOVO $\eta^2 = 2.5 \times 10^{-3}$ (LOVO/internal ratio 6.0), reflecting estimation on the pooled within-fold sample; ComBat-joint remained at 4.4×10^{-4} (ratio 0.9). CovBat showed moderate degradation ($\eta^2 = 1.0 \times 10^{-3}$, ratio 4.0). RELIEF and LME/FE failed external validation (η^2_{LOVO}: 0.19 and 0.20; PERMANOVA $p < 0.01$ in all folds; ratios 428 and 253; Figure 4a–d). This pattern is explained by structural constraints: RELIEF's ICA unmixing matrix is non-transferable across vendor sets, and LME/FE cannot estimate offsets for unseen vendors. Notably, CovBat and RELIEF—the two methods that project the held-out vendor onto the training PCA/ICA basis—showed test-vendor pre/post correlations below 1 (fold-wise range $r = 0.85$–0.97; fold means 0.90 for CovBat and 0.95 for RELIEF). The affine z-score methods, by contrast, preserved held-out-vendor subject rankings exactly by construction.

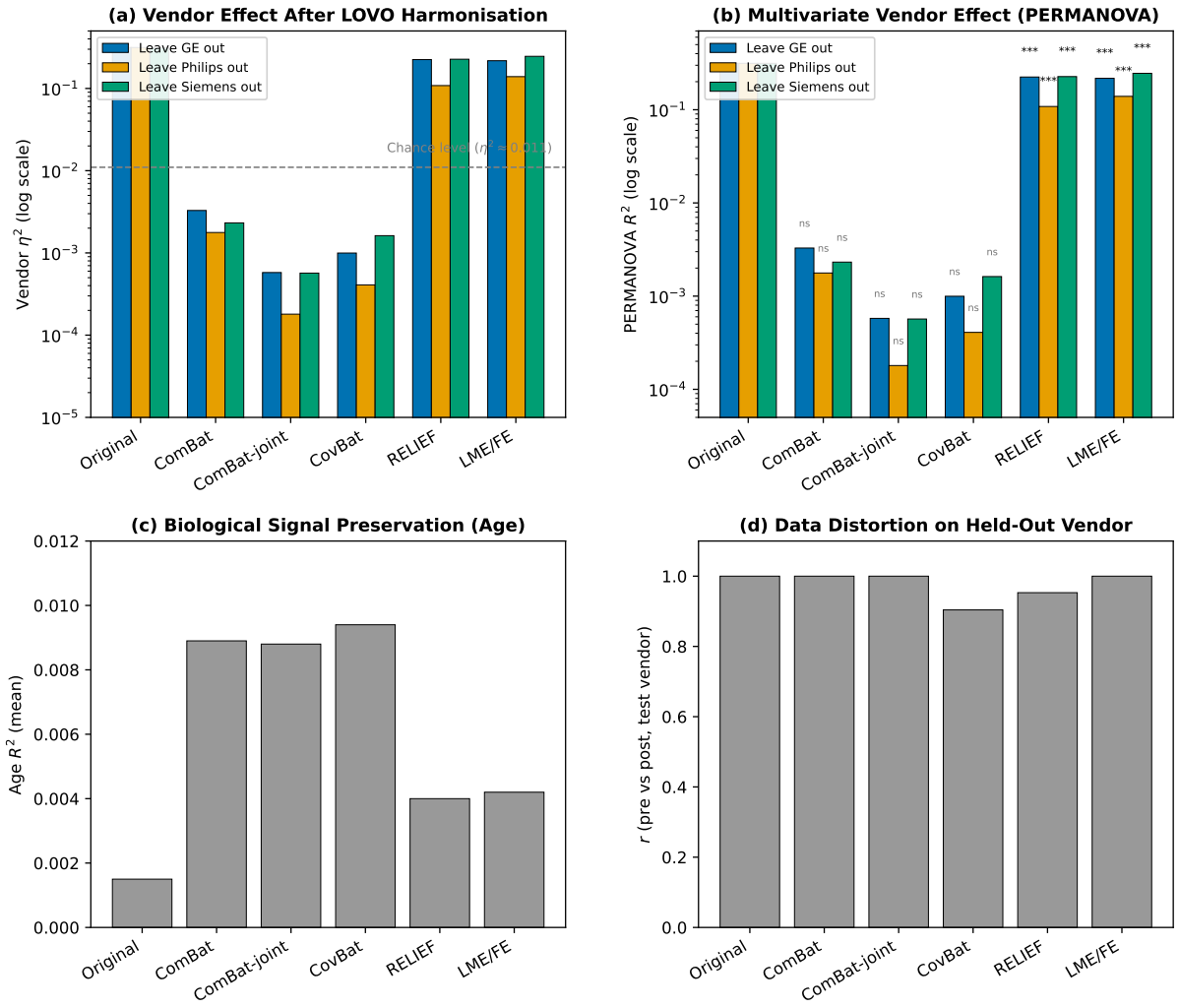


Figure 4: Leave-one-vendor-out (LOVO) external validation. **(a)** Vendor effect size (η^2, eta-squared from one-way ANOVA) by method and held-out vendor, shown on logarithmic scale. **(b)** Multivariate vendor structure (PERMANOVA R^2, permutational multivariate analysis of variance) with significance indicators (asterisks denote $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; ns = not significant). **(c)** Age-related biological signal (semi-partial R^2) recovery. **(d)** Subject-level preservation (Pearson $r_{\text{pre,post}}$) on unseen vendors. PERMANOVA R^2 equals the mean per-metric η^2 under the z-scored Euclidean distance (Methods), so panels (a) and (b) are numerically identical except for the significance annotation. For the unharmonised baseline (Original) no transformation is fitted, so its pooled within-fold R^2 (0.315) is identical across folds by construction.

3.7 Structural Constraints

Three methodological observations converge on a single structural limitation of the dataset: only 3 vendors, with sites perfectly nested (43 sites in the dataset, 40 in the analysis sample). The LME random-effects covariance matrix was singular in 56% of cases because 3 vendor levels provide insufficient information for stable REML variance-component estimation ($k \geq 5$ recommended; Table 3). RELIEF's ICA found no vendor-associated components because ComBat-joint preprocessing had already removed

vendor mean/scale effects, leaving insufficient residual vendor structure for ICA to detect. These constraints are inherent to the retrospective design; they do not invalidate the internal benchmark findings but bound their generalisability to similarly structured consortia.

Table 3: LME convergence status by biomarker and cohort (complete-case unified fits). FE = fixed-effects fallback (singular random-effects covariance matrix); LME = converged linear mixed-effects model. LME converged in 7 of 16 combinations (44%); only MD and AD achieved stable LME fits in both cohorts. All 9 FE-fallback cases produced post-harmonisation η^2 reductions exceeding 99%, confirming that FE centring is sufficient for vendor-effect removal even when REML fails. The single converged-LME outlier (HC, T2w-CSA; post $\eta^2 = 4.2 \times 10^{-3}$) reflects an unstable variance-component estimate rather than FE fallback.

Cohort	Biomarker	N	η^2 (pre)	η^2 (post)	Status
HC	T2w-CSA	188	0.0530	4.19×10^{-3}	**LME**
	GM-CSA	188	0.1761	5.09×10^{-4}	FE
	MTR	188	0.2052	1.01×10^{-4}	FE
	MTsat	188	0.3353	4.78×10^{-5}	FE
	FA	188	0.2902	1.28×10^{-4}	FE
	MD	188	0.4646	3.21×10^{-4}	**LME**
	AD	188	0.7003	5.33×10^{-4}	**LME**
	RD	188	0.2927	5.07×10^{-4}	FE
ALL	T2w-CSA	246	0.0441	3.65×10^{-5}	FE
	GM-CSA	246	0.1673	2.06×10^{-5}	FE
	MTR	246	0.2428	2.07×10^{-5}	FE
	MTsat	246	0.3596	1.53×10^{-4}	**LME**
	FA	246	0.3116	3.55×10^{-4}	**LME**
	MD	246	0.5064	2.88×10^{-4}	**LME**
	AD	246	0.7362	4.68×10^{-4}	**LME**
	RD	246	0.3199	4.72×10^{-4}	FE

4 Discussion

This study provides, to our knowledge, the first systematic benchmark of statistical harmonisation methods for multi-vendor spinal cord qMRI biomarkers. Six principal findings emerge.

First, baseline vendor effects in spinal cord qMRI are large ($\eta^2 = 0.31$, PERMANOVA $R^2 = 0.315$)—comparable to or exceeding brain imaging reports [9–11]—and dominate biological variation by approximately two orders of magnitude. In absolute terms, the

observed baseline corresponds to 31% of cross-biomarker variance attributable to vendor. This suggests that vendor offsets, if uncontrolled, would confound biomarker contrasts across vendors in any pooled analysis. Physical harmonisation, such as vendor-specific recalibration of magnetisation transfer metrics [7], offers a complementary acquisition-level route but requires prospective protocol control; statistical post-hoc harmonisation, as benchmarked here, applies retrospectively to existing consortia data. These results motivate harmonisation as a candidate preprocessing step for multi-centre spinal cord studies of similar structure (three vendors, perfectly nested sites), a need echoed by recent European quality initiatives [6] and by multicentre MRI harmonisation recommendations [21].

Second, all five methods reduce vendor effects to statistical negligibility on the variance-decomposition metrics each method explicitly optimises (η^2, PERMANOVA), whereas leakage-free machine-learnability—a criterion no method optimises directly—reveals substantial but incomplete removal (AUC 0.63–0.81 across methods, pathology-inclusive cohort; HC 0.67–0.85; versus 0.99 unharmonised and 0.50 chance). We note that evaluating vendor detectability after full-sample harmonisation, as is common practice, is optimistically biased because the harmonisation parameters absorb test-fold structure; our fold-internal protocol (Methods) quantifies the deployment-relevant residual. The choice among methods therefore hinges not on whether they work, but on the downstream analytic goal. For subject-level inference (individual prediction, longitudinal change detection), LME/FE or ComBat-joint may be appropriate. For pooled-group analyses where maximal multivariate vendor removal is the priority, CovBat achieved the lowest residual vendor AUC (0.629 ALL), albeit at the cost of the lowest subject-level preservation ($r = 0.782$ HC); RELIEF was intermediate. Our machine-learnability results quantify this hierarchy directly in spinal cord data: despite reducing vendor η^2 to between 2.5×10^{-4} and 7.9×10^{-4}, LME/FE left the most detectable residual signature (AUC 0.81), whereas CovBat retained the least (0.63). The residual detectability of ComBat-harmonised profiles (AUC 0.71) is in the range previously reported for location-scale harmonisation in multi-site brain imaging [11], although direct numerical comparison across modalities is constrained by differences in feature space, sample size, and number of sites.

Third, the Pareto frontier between decontamination and preservation is reproducible across cohorts and persisted under the reduced-imbalance and true-equal-n subsample analyses (Design C; Section 3.5). The higher baseline vendor-variance fractions of these subsamples ($\eta^2 = 0.374$ and 0.389 vs. 0.315) imply correspondingly lower pooled-r ceilings ($\sqrt{1-\eta^2} \approx 0.79$ and 0.78), which accounts for their lower absolute r values, indicating that the preservation hierarchy tracks correction depth rather than the dataset's vendor imbalance. The frontier should nonetheless be read as a descriptive characterisation of this dataset's structure rather than a universal method ranking, and the decontamination–

preservation trade-off it visualises remains a useful framing for method selection.

Fourth, LOVO external validation differentiates methods sharply: ComBat and ComBat-joint generalise to unseen vendors without significant degradation, while RELIEF and LME/FE fail. Held-out-vendor degradation was markedly smaller for ComBat-joint than for per-metric ComBat (LOVO/internal η^2 ratio 0.9 vs. 6.0), consistent with joint empirical Bayes estimation sharing statistical strength across biomarkers when vendor offsets must be estimated from only two training vendors. For consortia anticipating addition of new vendor scanners, our LOVO results suggest ComBat-joint as a candidate method, with the caveat that this is a single-dataset observation. Out-of-sample harmonisation for new sites is an active methodological frontier: ComBat-Predict provides a principled framework for applying trained harmonisation models to unseen sites without re-accessing reference data [16], and our LOVO results independently support ComBat-family—in particular joint—harmonisation for this deployment scenario.

Fifth, the structural constraints revealed by this benchmark—vendor-site nesting, LME singular covariance in 56% of cases, and RELIEF ICA inactivation—reflect the 3-vendor, perfectly-nested spine-generic design. These are not methodological failures but intrinsic dataset properties that bound interpretability: reported vendor effects cannot be partitioned into vendor-level versus site-level components, and LME/RELIEF's degradation in low-vendor-count settings is expected given their modelling assumptions. Larger consortia with more vendors and crossed designs would be expected to relax these constraints, by providing enough vendor levels for stable REML and richer residual structure for ICA.

Sixth, biological-signal amplification after harmonisation was selective and interpretable: age associations absent at baseline (all $p \geq 0.19$) became significant in the diffusion metrics (Section 3.4; mean 4.8–5.0-fold across cohorts), and pathology classification improved (AUC 0.52 to 0.62). This demonstrates that harmonisation does not merely remove noise—it unmasks signal previously obscured by vendor variance. The concurrent pathology-classification gain mirrors the disease-classification gains reported after harmonisation in brain imaging [11], extending that observation to spinal cord biomarkers.

Limitations. (i) Vendor-site perfect nesting precludes decomposition of vendor-level versus site-level effects; our findings reflect the composite "vendor-in-site" effect relevant to practical consortia. (ii) The dataset covers only three vendors with substantial imbalance (Siemens: 166/246, 67% of the ALL cohort; 130/188, 69% of HC); the reduced-imbalance and true-equal-n subsample analyses retained the preservation hierarchy (Section 3.5), but extrapolation to other manufacturers and more balanced consortia requires caution. (iii) LME converged in only 7 of 16 biomarker×cohort combinations (44%); the reported LME/FE preservation advantage ($r = 0.82$) may reflect the minimal correction of the FE

fallback rather than LME shrinkage.

(iv) RELIEF's ICA inactivation ($q \geq 0.99$ for all components) may be dataset-specific; Zhang et al. [12] demonstrated RELIEF's effectiveness in a three-site DTI study where latent vendor structure persisted after ComBat preprocessing—suggesting that consortia with stronger residual vendor-covariance structure may benefit from RELIEF, whereas our 43-site design with ComBat-joint preprocessing may have already exhausted this structure. (v) LOVO validation used z-score matching for unseen-vendor transformation—the standard ComBat-from-template approach [9]—but does not test prospective acquisition on genuinely new hardware; principled out-of-sample frameworks such as ComBat-Predict [16] offer a natural extension for prospective new-site harmonisation.

(vi) Fold-internal machine-learnability assumes vendor labels are observed at deployment (the standard assumption of ComBat-style harmonisation); residual vendor detectability remained above chance for every method (AUC up to 0.85 for LME/FE), so harmonised pooling is not vendor-blind, and pooled analyses should continue to report vendor composition. (vii) This analysis was not pre-registered; however, all harmonisation methods were selected and evaluated on a pre-specified endpoint framework prior to analysis, and the complete analytical pipeline is publicly documented in the accompanying code repository, ensuring full computational reproducibility.

Conclusion. Statistical harmonisation reduces vendor effects to statistical negligibility on variance-decomposition metrics and removes the majority—though not all—of the machine-detectable vendor signature in multi-centre spinal cord qMRI. It may be appropriate as a default preprocessing step for datasets of similar structure (three vendors, perfectly nested sites), with residual vendor-related confounding reported alongside pooled analyses. ComBat-joint provides the best overall balance in our sample: within-sample vendor removal was comparable to per-metric ComBat ($\eta^2 = 4.9$ vs. 4.1×10^{-4}, both far below the chance level) and the two methods were indistinguishable on leakage-free vendor detectability (AUC 0.704 vs. 0.706 in ALL), yet ComBat-joint degraded markedly less on unseen manufacturers (LOVO/internal ratio 0.9 vs. 6.0) and preserved subject-level information slightly better ($r = 0.806$ vs. 0.789 in HC). ComBat-joint may therefore be appropriate for broad multi-centre deployment; per-metric ComBat remains appropriate when individual biomarkers are harmonised in isolation.

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